

1 Latent Relational Source Generation

Inputs

- Domain
- Attribute count
- Attribute cardinality
- Past table names

Schema generation

```
{
  "table_name": "...",
  "attributes": [...],
  "verbose_headers": [...],
  "ranges": [...],
  "value_col": "...",
  "unit": "...",
  "decimals": ...
}
```

Latent relational source

Company	Facility	...	Value
X	Atlanta	...	42.1
Y	Busan	...	31.4
Z	London	...	19.2
...	...	...	...

SQL query provenance

2 SQL Supervision

Parallel

one scalar per table, then aggregate

```
-- T1:SELECT value FROM R WHERE company='X'AND sector='AI'→ a1
-- T2:SELECT value FROM R WHERE company='Y'AND sector='AI'→ a2
-- T3:SELECT value FROM R WHERE company='Z'AND sector='AI'→ a3
y = g(a1, a2, a3), g ∈ {SUM, AVG, MAX, MIN}
```

Sequential

output of step i becomes lookup key of step i+1

```
-- T1:SELECT company FROM CEOs WHERE ceo='John Doe'→ b1
-- T2:SELECT facility FROM Assign WHERE company=b1→ b2
-- T3:SELECT value FROM R&D WHERE facility=b2
      AND sector IN ('AI', 'Finance') → {a1, a2}
y = g(a1, a2), applied over the last view's outputs
```

Gold label

3 Rendering Non-Relational Views

Parallel views: each view contributes one independent scalar

Company X		
2023 indicators per employee	Branch location	
	UK	US
Weekly water usage 2022	5.89 gallons	7.16 gallons
Weekly water usage 2023	5.60 gallons	--
Perturbations: Pivot + unit change + n.a.		

Company Y		
Branch locations 2023	Weekly liters (per employee)	
	Water	Oil
France	22	31
UK	21.5	not available
Perturbations: Pivot + unit in header + n.a.		

Company Z			
Weekly usage per employee		Rbnach citise	
		Lodnon (UK)	Tokoy (Jaapn)
Waetr	2022	0.022 m ³	0.031 m ³
	2023	0.024 m ³	0.028 m ³
Perturbations: Row/col pivot + unit change + OCR			

Sequential views: each lookup key parameterizes the next view

CEO names			Facility assignment			R&D expenses		
Year	Compaines		Facilities	Compaines		Costs (2023)	Investments (millions)	
	X	Y		X	Y		AI Research	Finance Research
2022	John Doe	Lee Seo-yeon	Primary	Atlanta Lab	Seoul Lab	Atlanta Lab	42.1	38.9
2023	Jnae Mya	n.a.	Backup	n.a.	Busan Hub	Seoul Lab	53.8 - as written in [1]	n.a.
Perturbations: Pivot + n.a. + OCR			Perturbations: Pivot + n.a. + empty column			Perturbations: Row/Col pivot + text insertion		

Transformations

Layout edits

- Pivot (hierarchical headers)
- Column merging
- Blank row/column insertion
- Attribute name change

Cell edits

- Typographical noise
- Footnote / text insertion
- Null / N.A. insertion

Numerical edits

- Unit-of-measurement change

4 Question Verbalization & Verification

Question Generation

Question Verification

Natural Language Question

Tables

parallel non-relational tables

sequential non-relational tables

Natural Language Questions

What is the average water consumption in liters of the top 3 S&P 500 companies in 2023, considering only UK branches?

What is the average 2023 investment of the company where John Doe is CEO? Restrict to AI and the financial sector.

Answers

22.23

40.5